

Section 3: System Design Fundamentals

Protocols

This is section 3 of the course as a part of system design fundamentals. In this session we will talk about protocols, explore the core communication protocols that power modern system architecture from TCP and UDP to HTTP, REST, WebSockets and beyond. We will break down how these protocols work, when to use them, and why they matter in the real world system design.

By the end of this session, you will have a solid understanding of the protocols that drive our web applications, APIs, and real-time communications.

Section Agenda

- Lecture 1: TCP & UDP - Understanding the Basics
- Lecture 2: HTTP - The backbone of the web
- Lecture 3: REST & RESTfulness - API Design Principles
- Lecture 4: Real-Time Communication Protocols
- Lecture 5: Modern API Protocols - Beyond REST
- Lecture 6: Summary & Practical Applications

This section will give you a strong foundation in protocols essential for system design.